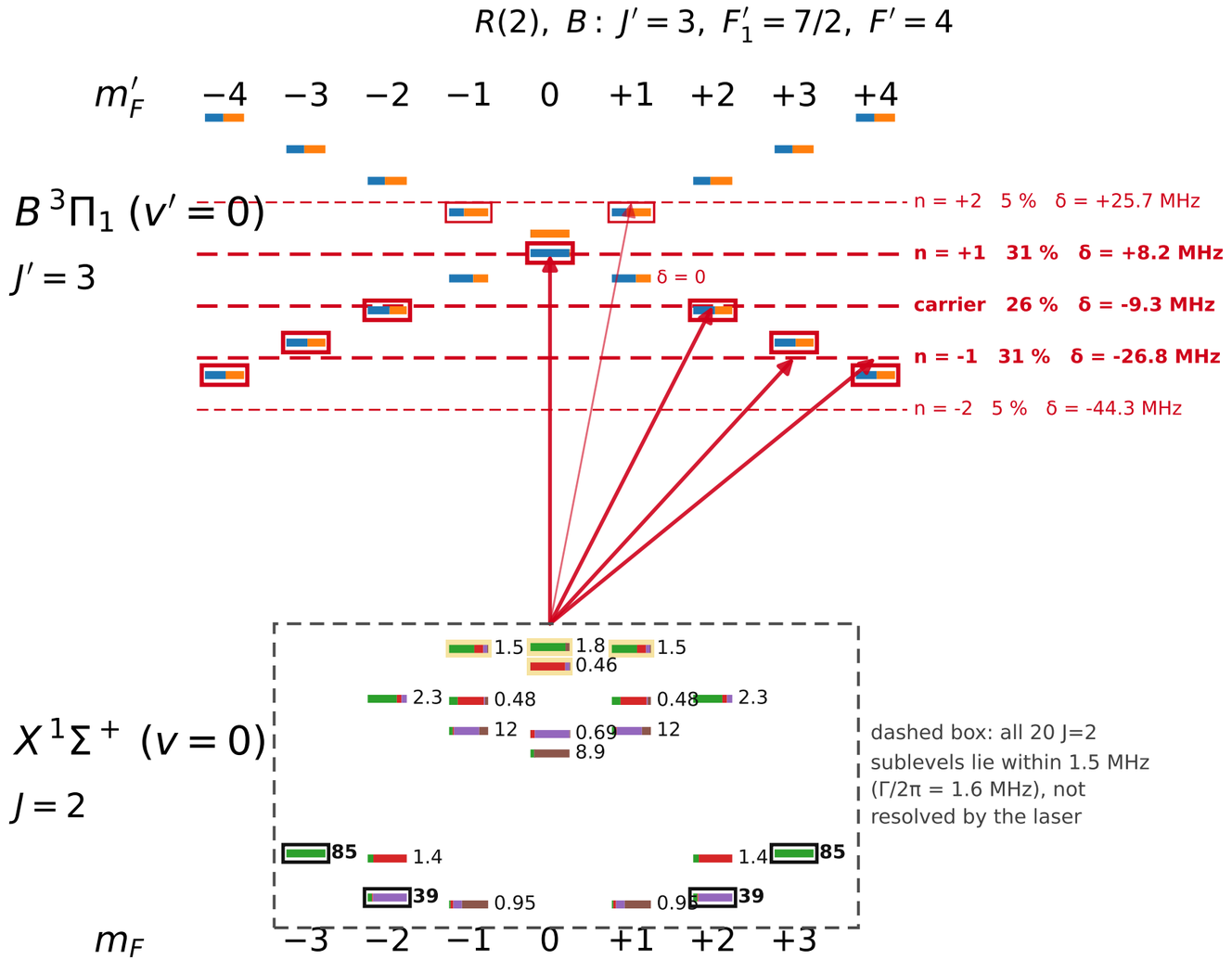
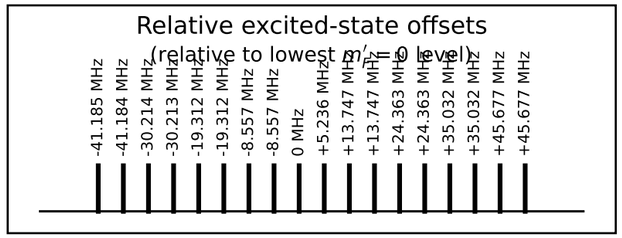
CU2 beam 1 | R(2) $F_1' = 7/2$, $F' = 4$ | E = 200 V/cm | B = 1 mG (model)



dashed box: all 20 J=2 sublevels lie within 1.5 MHz (Γ/2π = 1.6 MHz), not resolved by the laser



Opposite-parity-parent separation at 200 V/cm
- $m_F' = -4$: 86.861 MHz
- $m_F' = -3$: 65.245 MHz
- $m_F' = -2$: 43.675 MHz
- $m_F' = -1$: 22.304 MHz
- $m_F' = 0$: 5.236 MHz
- $m_F' = +1$: 22.304 MHz
- $m_F' = +2$: 43.675 MHz
- $m_F' = +3$: 65.245 MHz
- $m_F' = +4$: 86.861 MHz

Zero-field parity splitting
5.248 MHz

Parity mixing at 200 V/cm (lower level)
- $m_F' = -4$: 53.0/47.0
- $m_F' = -3$: 54.0/46.0
- $m_F' = -2$: 56.0/44.0
- $m_F' = -1$: 61.7/38.3
- $m_F' = 0$: 100.0/0.0
- $m_F' = +1$: 61.7/38.3
- $m_F' = +2$: 56.0/44.0
- $m_F' = +3$: 54.0/46.0
- $m_F' = +4$: 53.0/47.0

zero-field $P' = -1$ character
zero-field $P' = +1$ character

Ground-state zero-field hyperfine character
$F_1 = 5/2,\ F = 3$
$F_1 = 5/2,\ F = 2$
$F_1 = 3/2,\ F = 2$
$F_1 = 3/2,\ F = 1$

Dashed lines and arrows: CU2 beam 1 at setting R2, carrier and sidebands n of the 17.5 MHz EOM (beta = 1.5), % = power fraction J_n(beta)^2. Drawn for the main ground level |F1=5/2, F=3, mF=+0>; delta = detuning from its resonance with |F1'=7/2, F'=4, mF'=+1, P'=-> (marked delta = 0).
For the other ground levels the lines move by -1.5 to +0.0 MHz. Coloured outline and arrow: sublevels a line reaches, i.e. within 4 Gamma/2pi = 6.4 MHz and with addressed-parity character > 5 % (line minus level): n=-2: none; n=-1 -> m'=±3 (P'-) -5.2 MHz and m'=±4 (P'-) +5.8 MHz; carrier -> m'=±2 (P'-) +1.4 MHz; n=+1 -> m'=0 (P'-) -0.4 MHz; n=+2 -> m'=±1 (P'+) +3.4 MHz.
Number right of each J=2 level: population left in J=2 after CU2 for a molecule starting in that level, in units of 1e-4 (setting R2: delta0 = 11.5 MHz, f_AOM = 191.0 MHz; 80 + 24 mW, 2.4 cm beam).
Average over the 20 levels: 14.9e-4.
Bold black outline: residual more than twice that average. Yellow band: SPB-related level (J=2, m_J=0 content > 0.5; SPB transfers out of these, what it leaves CU2 must remove).
Single trajectory on the beam axis, no transverse velocity; 13 passes, 10 % loss per pass, X/Z switching at 0.5 MHz.
Beam 0 (F'=3, delta0 = +11.5 MHz) lies about 200 MHz lower and is drawn in the F'=3 diagram. At R1 (delta1 = +8.17 MHz) the beam-1 lines sit at the same frequencies, each order shifted by one.
At setting R1 (delta0 = 11.5 MHz, f_AOM = 208.5 MHz) the same three worst levels give 131 (F1=5/2, F=3, mF=±3), 19 (F1=3/2, F=2, mF=±2), 12 (F1=3/2, F=2, mF=±1); average 17.6e-4.